Arduino Code:

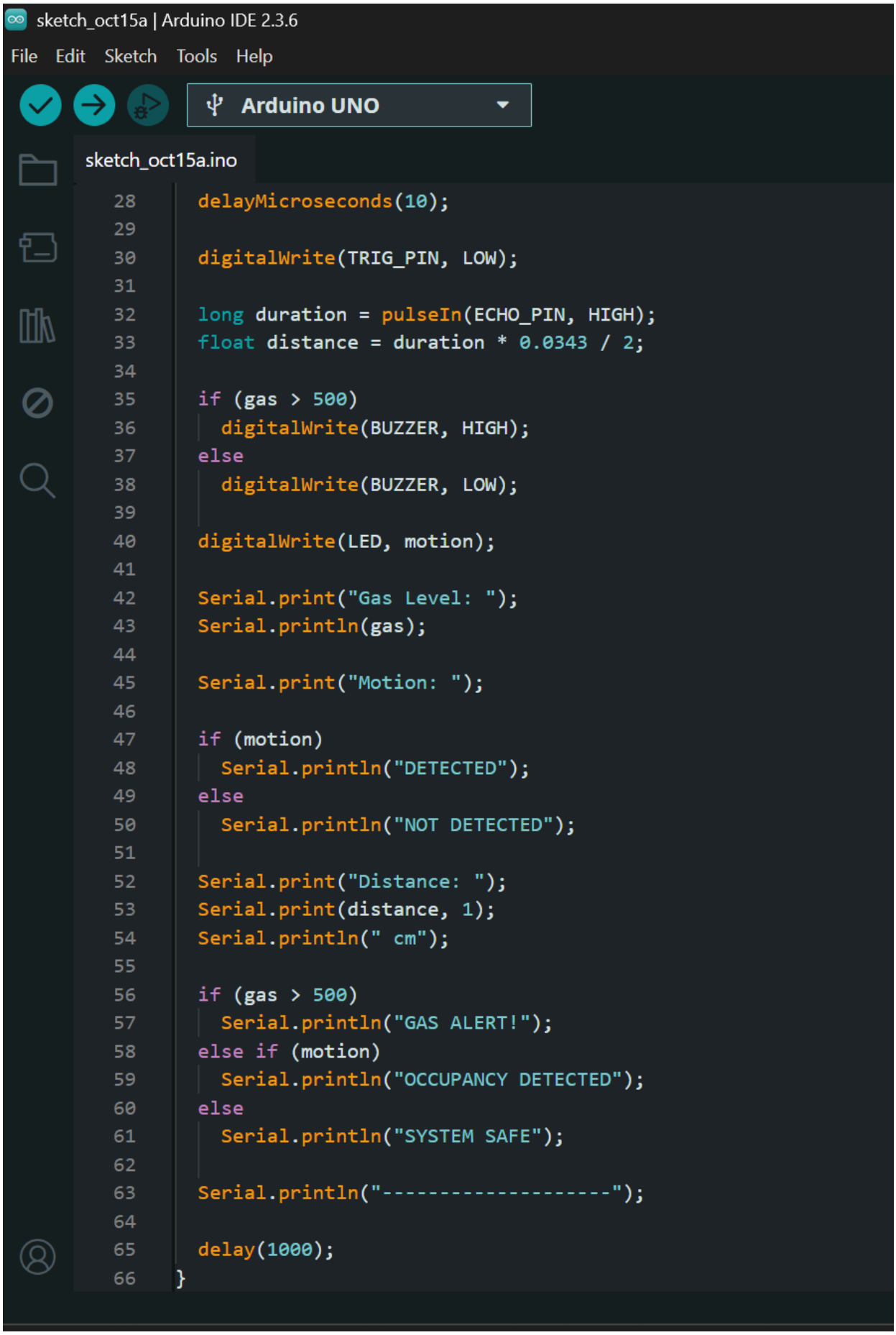
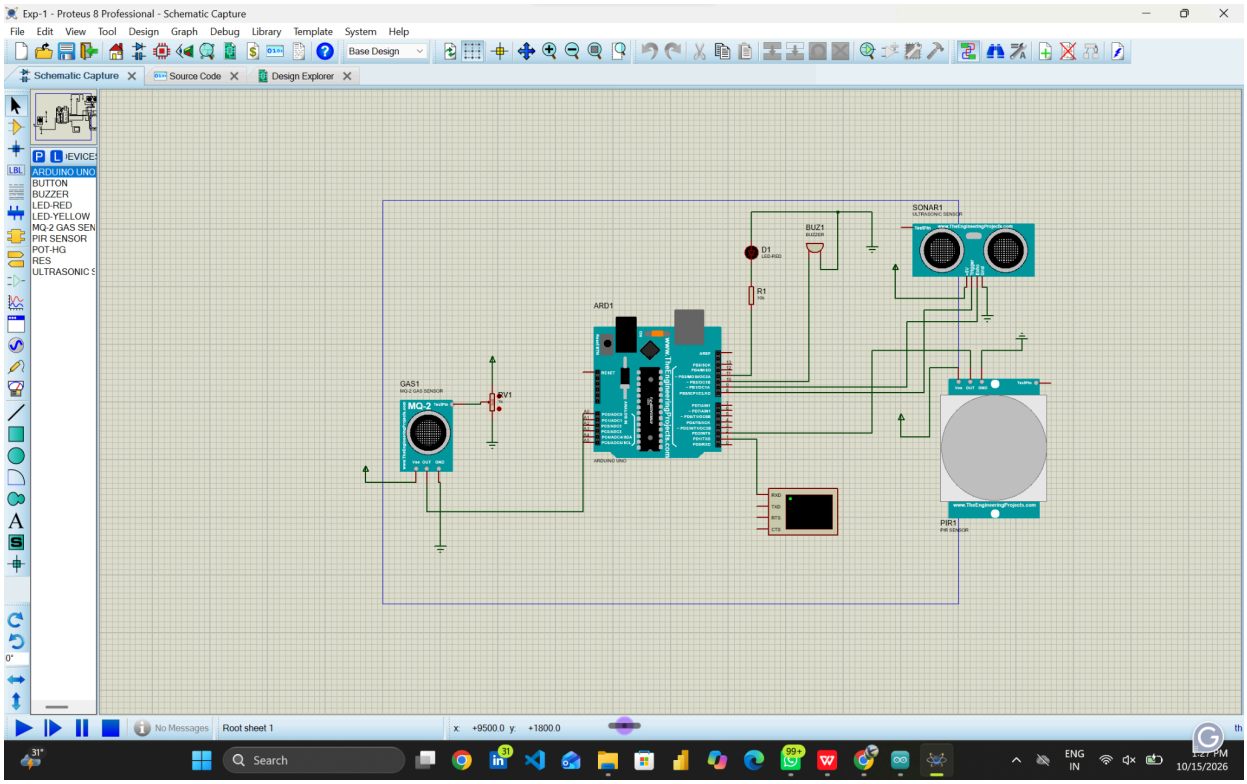
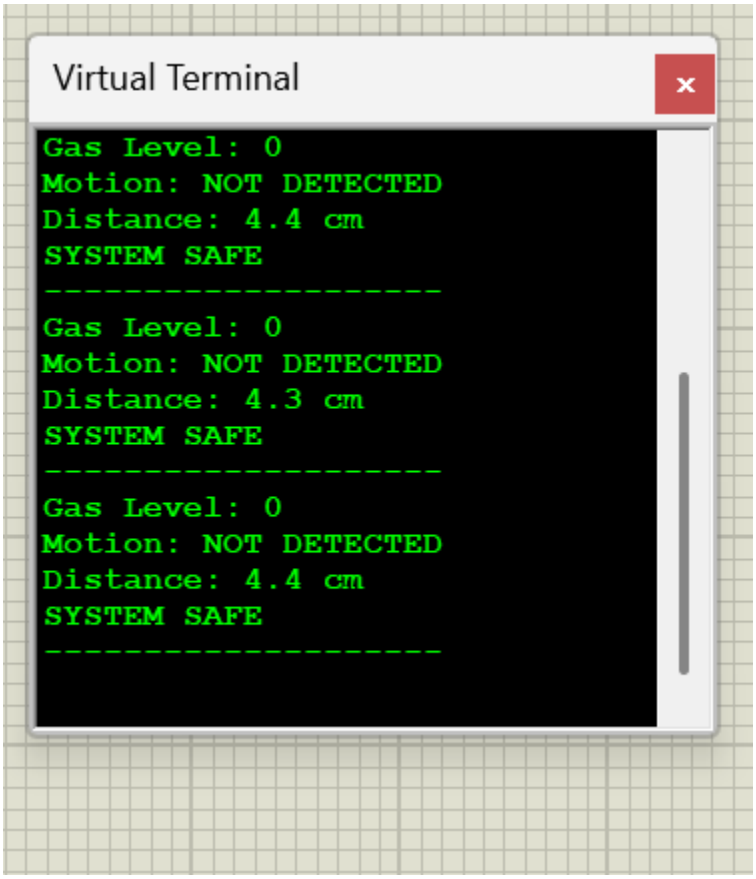

sketch_oct15a | Arduino IDE 2.3.6
File Edit Sketch Tools Help
Arduino UNO
sketch_oct15a.ino
28 delayMicroseconds(10);
29
30 digitalWrite(TRIG_PIN, LOW);
31
32 long duration = pulseIn(ECHO_PIN, HIGH);
33 float distance = duration * 0.0343 / 2;
34
35 if (gas > 500)
36 digitalWrite(BUZZER, HIGH);
37 else
38 digitalWrite(BUZZER, LOW);
39
40 digitalWrite(LED, motion);
41
42 Serial.print("Gas Level: ");
43 Serial.println(gas);
44
45 Serial.print("Motion: ");
46
47 if (motion)
48 Serial.println("DETECTED");
49 else
50 Serial.println("NOT DETECTED");
51
52 Serial.print("Distance: ");
53 Serial.print(distance, 1);
54 Serial.println(" cm");
55
56 if (gas > 500)
57 Serial.println("GAS ALERT!");
58 else if (motion)
59 Serial.println("OCCUPANCY DETECTED");
60 else
61 Serial.println("SYSTEM SAFE");
62
63 Serial.println("--------------------");
64
65 delay(1000);
66 }

Diagram:


Exp-1 - Proteus 8 Professional - Schematic Capture
File Edit View Tool Design Graph Debug Library Template System Help
Schematic Capture
Source Code
Design Explorer
ARDUINO UNO
BUTTON
BUZZER
LED-RED
LED-YELLOW
MQ-2 GAS SEN
PIR SENSOR
POT-HG
RES
ULTRASONIC
ARD1
GAS1
MQ-2
RV1
D1
R1
BUZ1
SONAR1
PIR1
Root sheet 1
No Messages
Search
ENG
IN
10/15/2026

Output:


Virtual Terminal
Gas Level: 0
Motion: NOT DETECTED
Distance: 4.4 cm
SYSTEM SAFE

Gas Level: 0
Motion: NOT DETECTED
Distance: 4.3 cm
SYSTEM SAFE

Gas Level: 0
Motion: NOT DETECTED
Distance: 4.4 cm
SYSTEM SAFE
